2.2 Complex Analysis: Solutions

Warm-up

1.

(a) $\bar{z} = x - iy$, so $u = x$, $v = -y$, and $u_x = 1 \neq -1 = v_y$ everywhere. Cauchy–Riemann fails at every point, so $\bar{z}$ is nowhere complex differentiable. (Directly: $\frac{\overline{z+h}-\bar{z}}{h} = \bar{h}/h$, which is 1 for real h and -1 for imaginary h.)

(b) At 0: $\dfrac{g(h)-g(0)}{h} = \dfrac{h\bar{h}}{h} = \bar{h} \to 0$, so $g'(0) = 0$. At $z_0 \neq 0$, $u = x^2 + y^2$, $v = 0$; CR requires $2x = 0$ and $2y = 0$, so it fails, and g is not complex differentiable there. Since g is differentiable only at the single point 0, it is not holomorphic on any open set.

2.

$u_{xx} + u_{yy} = 2 - 2 = 0$, so a harmonic conjugate can exist. From $v_y = u_x = 2x + 1$ we get $v = 2xy + y + \varphi(x)$. Then $v_x = 2y + \varphi'(x)$, and CR requires $v_x = -u_y = 2y$, so $\varphi' = 0$ and $\varphi = c$, a real constant. Thus

$$f = (x^2 - y^2 + x) + i(2xy + y + c) = z^2 + z + ic, \quad c \in \mathbb{R}.$$

These are all holomorphic, and any holomorphic f with real part u is of this form (two holomorphic functions with the same real part differ by an imaginary constant, by CR).

3.

(a) $|n^3|^{1/n} = e^{3 \ln n/n} \to 1$, so $R = 1$.

(b) The coefficient a_k is 1 if $k = n!$ for some $n \geq 1$, and 0 otherwise. (The value $k = 1$ arises from $n = 1$ only, since we start at $n \geq 1$.) So $\limsup |a_k|^{1/k} = 1$ and $R = 1$.

(c) With $a_n = (n!)^2/(2n)!$,

$$\frac{a_n}{a_{n+1}} = \frac{(2n+2)(2n+1)}{(n+1)^2} = \frac{2(2n+1)}{n+1} \to 4,$$

so by the ratio test $R = 4$.

4.

Parametrise $z = e^{it}$, $0 \leq t \leq 2\pi$, $dz = ie^{it}dt$. Then

$$\int_C \bar{z}\, dz = \int_0^{2\pi} e^{-it}\, ie^{it}\, dt = 2\pi i, \qquad \int_C z^n\, dz = i\int_0^{2\pi} e^{i(n+1)t}\, dt = \begin{cases} 2\pi i & n = -1, \\ 0 & n \neq -1. \end{cases}$$

Cauchy's theorem needs a holomorphic integrand, and $\bar{z}$ is not holomorphic (Problem 1). In fact on C, $\bar{z} = 1/z$, which explains the value.

5.

$$\frac{1}{z(z-1)} = -\frac{1}{z} \cdot \frac{1}{1-z}.$$

(a) For $0 < |z| < 1$: $\dfrac{1}{1-z} = \sum_{n\geq 0} z^n$, so

$$f(z) = -\sum_{n\geq 0} z^{n-1} = -\frac{1}{z} - 1 - z - z^2 - \cdots.$$

(b) For $|z| > 1$: $\dfrac{1}{z-1} = \dfrac{1}{z}\cdot\dfrac{1}{1-1/z} = \sum_{n\geq 0} z^{-n-1}$, so

$$f(z) = \frac{1}{z}\sum_{n\geq 0} z^{-n-1} = \sum_{n\geq 0} z^{-n-2} = \frac{1}{z^2} + \frac{1}{z^3} + \cdots.$$

The residue at 0 is read from the expansion valid in a punctured disc around 0, i.e. (a): $\operatorname{Res}(f, 0) = -1$. (The coefficient of z^{-1} in (b) is 0; this is not the residue at 0. It equals the sum of the residues at 0 and 1, namely $-1 + 1$.)

6.

For $z = x + iy$,

$$|z+i|^2 - |z-i|^2 = (x^2 + (y+1)^2) - (x^2 + (y-1)^2) = 4y.$$

So for $z \in \mathbb{H}$, $|z-i| < |z+i|$ and $|\varphi(z)| < 1$; also $z + i \neq 0$, so φ is holomorphic on $\mathbb{H}$. Solving $w = \frac{z-i}{z+i}$ gives $z = \psi(w) = i\frac{1+w}{1-w}$, holomorphic for $w \neq 1$. For $w \in \mathbb{D}$,

$$\operatorname{Im}\psi(w) = \operatorname{Re}\frac{1+w}{1-w} = \operatorname{Re}\frac{(1+w)(1-\bar{w})}{|1-w|^2} = \frac{1-|w|^2}{|1-w|^2} > 0,$$

(the cross terms $w - \bar{w}$ are purely imaginary). So ψ maps $\mathbb{D}$ into $\mathbb{H}$, and $\varphi\circ\psi$, $\psi\circ\varphi$ are the identity where defined. Hence $\varphi : \mathbb{H} \to \mathbb{D}$ is a holomorphic bijection with holomorphic inverse ψ.

Core

7.

Write $f(z) = \sum_{k\geq 0} a_k z^k$ (valid on all of $\mathbb{C}$). By the Cauchy estimates on the circle $|z| = R$,

$$|a_k| = \frac{|f^{(k)}(0)|}{k!} \leq \frac{A + BR^n}{R^k}.$$

For $k > n$ the right side tends to 0 as $R \to \infty$, so $a_k = 0$. Hence f is a polynomial of degree at most n.

8.

$g = e^f$ is entire and $|g| = e^{\operatorname{Re} f} \leq e^M$. By Liouville g is constant, so $0 = g' = f'e^f$. Since e^f never vanishes, $f' \equiv 0$, and f is constant (on the connected set $\mathbb{C}$).

9.

Let $f(z) = 1/(1+z^4)$. Its poles are the fourth roots of -1: $e^{i\pi/4}, e^{3i\pi/4}, e^{5i\pi/4}, e^{7i\pi/4}$, all simple. In $\mathbb{H}$ lie $\alpha = e^{i\pi/4}$ and $\beta = e^{3i\pi/4}$. At a simple zero c of $1 + z^4$, $\operatorname{Res}(f, c) = \dfrac{1}{4c^3} = \dfrac{c}{4c^4} = -\dfrac{c}{4}$. So the sum of residues in $\mathbb{H}$ is

$$-\frac{\alpha+\beta}{4} = -\frac{1}{4}\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\frac{i\sqrt{2}}{4}.$$

Let Γ_R be $[-R, R]$ followed by the upper semicircle C_R, with $R > 1$. Then $\int_{\Gamma_R} f = 2\pi i \cdot \left(-\frac{i\sqrt{2}}{4}\right) = \frac{\pi\sqrt{2}}{2}$. On C_R, $|f| \le 1/(R^4-1)$, so $\left|\int_{C_R} f\right| \le \pi R/(R^4-1) \to 0$. Since $\int_{-\infty}^{\infty} f$ converges absolutely,

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{\pi\sqrt{2}}{2} = \frac{\pi}{\sqrt{2}}.$$

10.

Put $z = e^{i\theta}$, so $\cos\theta = \frac{1}{2}(z + z^{-1})$ and $d\theta = dz/(iz)$:

$$\int_0^{2\pi} \frac{d\theta}{a + \cos\theta} = \int_{|z|=1} \frac{1}{a + \frac{1}{2}(z+z^{-1})} \cdot \frac{dz}{iz} = \int_{|z|=1} \frac{2\,dz}{i(z^2 + 2az + 1)}.$$

The roots of $z^2 + 2az + 1$ are $z_\pm = -a \pm \sqrt{a^2-1}$, real and distinct. Their product is 1 and $z_- < -1$, so $|z_+| < 1 < |z_-|$; only z_+ lies inside. The residue there is

$$\frac{2}{i(z_+ - z_-)} = \frac{2}{i \cdot 2\sqrt{a^2-1}} = \frac{1}{i\sqrt{a^2-1}}.$$

So the integral is $2\pi i \cdot \dfrac{1}{i\sqrt{a^2-1}} = \dfrac{2\pi}{\sqrt{a^2-1}}$.

11.

Let $f(z) = e^{iz}/(z^2+1)$. Its only pole in $\mathbb{H}$ is $z = i$ (simple), with residue $\dfrac{e^{i\cdot i}}{2i} = \dfrac{e^{-1}}{2i}$. With Γ_R as in Problem 9 ($R > 1$), $\int_{\Gamma_R} f = 2\pi i \cdot \frac{e^{-1}}{2i} = \frac{\pi}{e}$. On C_R, $z = x + iy$ with $y \ge 0$, so $|e^{iz}| = e^{-y} \le 1$ and $|f| \le 1/(R^2-1)$; hence $\left|\int_{C_R} f\right| \le \pi R/(R^2-1) \to 0$. Therefore $\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1}dx = \frac{\pi}{e}$. Taking real parts, $\int_{-\infty}^{\infty} \frac{\cos x}{x^2+1}dx = \frac{\pi}{e}$ (and the imaginary part, $\int \frac{\sin x}{x^2+1}dx$, is 0, as it must be by oddness).

12.

Let $f(z) = e^{iz}/z$ and $0 < \varepsilon < R$. Consider the closed contour made of: $[\varepsilon, R]$; the upper semicircle C_R from R to $-R$; $[-R, -\varepsilon]$; and the upper semicircle C_ε from $-\varepsilon$ to ε traversed clockwise. f is holomorphic inside and on this contour, so the total integral is 0.

The two segments. Substituting $x \mapsto -x$ on $[-R, -\varepsilon]$,

$$\int_{-R}^{-\varepsilon} \frac{e^{ix}}{x}dx + \int_{\varepsilon}^{R} \frac{e^{ix}}{x}dx = \int_{\varepsilon}^{R} \frac{e^{ix} - e^{-ix}}{x}dx = 2i\int_{\varepsilon}^{R} \frac{\sin x}{x}dx.$$

The large arc. With $z = Re^{i\theta}$, $|e^{iz}| = e^{-R\sin\theta}$ and $|dz/z| = d\theta$, so

$$\left|\int_{C_R} f\right| \le \int_0^{\pi} e^{-R\sin\theta}d\theta = 2\int_0^{\pi/2} e^{-R\sin\theta}d\theta \le 2\int_0^{\pi/2} e^{-2R\theta/\pi}d\theta < \frac{\pi}{R},$$

using $\sin\theta \ge 2\theta/\pi$ on $[0, \pi/2]$. This tends to 0.

The small arc. $f(z) = \frac{1}{z} + E(z)$ where $E(z) = (e^{iz}-1)/z$ is bounded near 0, say $|E| \le K$ for $|z| \le 1$. Then $\left|\int_{C_\varepsilon} E\right| \le \pi\varepsilon K \to 0$, and with $z = \varepsilon e^{i\theta}$, θ from π to 0, $\int_{C_\varepsilon} \frac{dz}{z} = \int_\pi^0 i\,d\theta = -i\pi$.

Letting $\varepsilon \to 0$ and $R \to \infty$: $2i \lim \int_\varepsilon^R \frac{\sin x}{x} dx - i\pi = 0$. Since $\frac{\sin x}{x}$ extends continuously to $x = 0$, the limit as $\varepsilon \to 0$ is harmless, and $\lim_{R\to\infty} \int_0^R \frac{\sin x}{x} dx = \frac{\pi}{2}$.

13.

(a) $\sin z = z\left(1 - \frac{z^2}{6} + O(z^4)\right)$, so

$$\frac{1}{z^2 \sin z} = \frac{1}{z^3}\left(1 + \frac{z^2}{6} + O(z^4)\right) = \frac{1}{z^3} + \frac{1}{6z} + O(z).$$

Pole of order 3, residue $\frac{1}{6}$.

(b) $e^{1/z} = \sum_{n\geq 0} \frac{1}{n! z^n}$ has infinitely many negative powers: essential singularity. Residue = coefficient of z^{-1} = 1.

(c) $\cot z = \cos z / \sin z$. At $z = k\pi$, sin has a simple zero ($\sin'(k\pi) = \cos k\pi = (-1)^k \neq 0$) and $\cos(k\pi) \neq 0$. So cot has a simple pole with residue $\dfrac{\cos k\pi}{\cos k\pi} = 1$.

14.

On $|z| = 2$: $|z^5| = 32$ and $|3z^3 + 7| \leq 24 + 7 = 31 < 32$. By Rouché, p and z^5 have the same number of zeros in $|z| < 2$, namely 5.

On $|z| = 1$: $|7| = 7$ and $|z^5 + 3z^3| \leq 4 < 7$. By Rouché, p and the constant 7 have the same number of zeros in $|z| < 1$, namely 0. Also p has no zeros on $|z| = 1$, since there $|p| \geq 7 - 4 > 0$.

So all 5 zeros lie in $1 < |z| < 2$.

15.

Let $f(z) = ze^{\lambda - z}$ and $g(z) = -1$. On $|z| = 1$, $|f(z)| = e^{\lambda - \operatorname{Re} z} \geq e^{\lambda - 1} > 1 = |g(z)|$. By Rouché, $f - 1$ has as many zeros in $\mathbb{D}$ as f, and f has exactly one (a simple zero at 0, as $e^{\lambda - z} \neq 0$). So $ze^{\lambda - z} = 1$ has exactly one solution in $\mathbb{D}$.

Now $h(x) = xe^{\lambda - x} - 1$ is continuous on $[0, 1]$ with $h(0) = -1 < 0$ and $h(1) = e^{\lambda - 1} - 1 > 0$. By the intermediate value theorem there is a root in $(0, 1)$, which lies in $\mathbb{D}$. By uniqueness, this is the solution.

16.

Both f and f^{-1} are holomorphic maps $\mathbb{D} \to \mathbb{D}$ fixing 0. By the Schwarz lemma, $|f(z)| \leq |z|$ and $|f^{-1}(w)| \leq |w|$. Applying the second with $w = f(z)$: $|z| \leq |f(z)|$. So $|f(z)| = |z|$ for all z, and the equality case of the Schwarz lemma gives $f(z) = e^{i\theta} z$.

17.

By the maximum modulus principle, $|f| \leq \max_{|z|=1} |f| = 1$ on $\overline{\mathbb{D}}$. Since f has no zeros in $\mathbb{D}$ and $|f| = 1$ on the circle, f has no zeros on $\overline{\mathbb{D}}$, so $1/f$ is holomorphic on $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$, with $|1/f| = 1$ on the circle. Maximum modulus again gives $|1/f| \leq 1$, i.e. $|f| \geq 1$. So $|f| \equiv 1$ on $\mathbb{D}$, and $|f|$ attains an interior maximum; by the maximum modulus principle f is constant.

Without the hypothesis $f \neq 0$: $f(z) = z$ satisfies everything else and is not constant.

18.

(a) No. Suppose such f exists. For even $n = 2k$, $f(1/(2k)) = 1/(2k)$. So $f(z) - z$ vanishes on $\{1/(2k)\}$, which has the limit point $0 \in \mathbb{D}$. By the identity theorem $f(z) = z$ on $\mathbb{D}$. But then $f(1/3) = 1/3 \neq -1/3$. Contradiction.

(b) $f(z) = z^2$ works. If f is any solution, $f(z) - z^2$ vanishes on $\{1/n : n \geq 2\}$, which accumulates at $0 \in \mathbb{D}$, so $f(z) = z^2$. The unique answer is $f(z) = z^2$.

19.

The zeros of f in $\overline{\mathbb{D}}$ are isolated (by the identity theorem, since $f \not\equiv 0$) and lie in a compact set, so there are finitely many; by hypothesis none are on the circle. Let $b_1, \dots, b_r$ be the distinct zeros in $\mathbb{D}$, with orders $m_1, \dots, m_r$. Near b_j, write $f(z) = (z - b_j)^{m_j} g(z)$ with g holomorphic and $g(b_j) \neq 0$. Then

$$\frac{f'(z)}{f(z)} = \frac{m_j}{z - b_j} + \frac{g'(z)}{g(z)}, \qquad \frac{zf'(z)}{f(z)} = \frac{m_j z}{z - b_j} + \frac{zg'(z)}{g(z)}.$$

The second term is holomorphic near b_j, and $\frac{m_j z}{z - b_j} = m_j + \frac{m_j b_j}{z - b_j}$, so $\operatorname{Res}\left(\frac{zf'}{f}, b_j\right) = m_j b_j$. The function zf'/f is holomorphic elsewhere on a neighbourhood of $\overline{\mathbb{D}}$ minus the b_j. By the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{zf'(z)}{f(z)} dz = \sum_j m_j b_j = a_1 + \cdots + a_k.$$

Challenge

20.

Let f be entire and injective; in particular non-constant. Consider $g(w) = f(1/w)$ on $\mathbb{C} \setminus \{0\}$, which has an isolated singularity at $w = 0$.

Not essential. By the open mapping theorem, $U = f(\mathbb{D})$ is a non-empty open set. If 0 were an essential singularity of g, then by Casorati–Weierstrass $g(\{0 < |w| < 1\}) = f(\{|z| > 1\})$ would be dense in $\mathbb{C}$, so it would meet U: there would be $|z| > 1$ and $|z'| < 1$ with $f(z) = f(z')$, contradicting injectivity.

Not removable. Then g would be bounded near 0, i.e. f bounded on $|z| > R$ for some R; f is also bounded on the compact set $|z| \leq R$. By Liouville f would be constant: contradiction.

So 0 is a pole of g, of some order $n \geq 1$. Write $f(z) = \sum_{k \geq 0} a_k z^k$ for all z. Then $g(w) = \sum_{k \geq 0} a_k w^{-k}$ for $w \neq 0$, which is the Laurent series of g on $0 < |w| < \infty$. By uniqueness of Laurent series and the definition of a pole of order n, $a_k = 0$ for $k > n$ and $a_n \neq 0$. So f is a polynomial of degree n.

Degree 1. Suppose $n \geq 2$. Then f' is a non-zero polynomial of degree $n - 1$, with finitely many roots. For each $c \in \mathbb{C}$, $f(z) - c$ has degree n and, by injectivity, exactly one root w_c, so $f(z) - c = a_n(z - w_c)^n$. Differentiating, $f'(z) = n a_n (z - w_c)^{n-1}$, so $f'(w_c) = 0$ (as $n - 1 \geq 1$). The points w_c are distinct for distinct c (since $f(w_c) = c$), so f' has infinitely many roots: contradiction. Hence $n = 1$ and $f(z) = az + b$ with $a \neq 0$.

21.

Use the branch $z^{a-1} = e^{(a-1)\log z}$ with $\log z = \ln|z| + i \arg z$, $\arg z \in (0, 2\pi)$, holomorphic on $\mathbb{C} \setminus [0, \infty)$. Let $f(z) = \dfrac{z^{a-1}}{1 + z}$. For $0 < \varepsilon < 1 < R$, take the keyhole contour: along the upper

side of the cut from ε to R; the circle $|z| = R$ anticlockwise; along the lower side of the cut from R to ε; the circle $|z| = \varepsilon$ clockwise. (Formally, use the limits of contours that stay a small angle δ away from the cut and let $\delta \to 0$; f extends continuously to each side of the cut.)

Values on the cut. On the upper side, $\arg z \to 0$, so $z^{a-1} = x^{a-1}$. On the lower side, $\arg z \to 2\pi$, so $z^{a-1} = x^{a-1}e^{2\pi i(a-1)} = x^{a-1}e^{2\pi ia}$.

Residue. The only singularity inside is the simple pole $z = -1 = e^{i\pi}$, with residue $(-1)^{a-1} = e^{i\pi(a-1)} = -e^{i\pi a}$.

Circles. On $|z| = R$: $|f| \le \dfrac{R^{a-1}}{R-1}$, so the integral is at most $\dfrac{2\pi R^a}{R-1} \to 0$ as $R \to \infty$ (since $a < 1$). On $|z| = \varepsilon$: $|f| \le \dfrac{\varepsilon^{a-1}}{1-\varepsilon}$, so the integral is at most $\dfrac{2\pi\varepsilon^a}{1-\varepsilon} \to 0$ as $\varepsilon \to 0$ (since $a > 0$).

Conclusion. With $I = \int_0^\infty \frac{x^{a-1}}{1+x}dx$ (convergent since the integrand is $O(x^{a-1})$ at 0 and $O(x^{a-2})$ at ∞), letting $\varepsilon \to 0$, $R \to \infty$:

$$I - e^{2\pi ia}I = 2\pi i \cdot (-e^{i\pi a}).$$

Since $1 - e^{2\pi ia} = e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) = -2i\sin(\pi a)\, e^{i\pi a}$,

$$I = \frac{-2\pi i\, e^{i\pi a}}{-2i\sin(\pi a)\, e^{i\pi a}} = \frac{\pi}{\sin(\pi a)}.$$

22.

(a) For $|z| < 1/|a|$ (all z if $a = 0$), $|\bar a z| < 1$, so $1 - \bar a z \neq 0$; since $1/|a| > 1$, φ_a is holomorphic on a neighbourhood of $\overline{\mathbb{D}}$. Compute

$$|1 - \bar a z|^2 - |z - a|^2 = 1 + |a|^2|z|^2 - |z|^2 - |a|^2 = (1 - |a|^2)(1 - |z|^2),$$

(the cross terms $-\bar a z - a\bar z$ appear in both and cancel). For $|z| < 1$ this is positive, so $|\varphi_a(z)| < 1$. Next, with $w = \varphi_a(z)$,

$$\varphi_{-a}(w) = \frac{w + a}{1 + \bar a w} = \frac{(z - a) + a(1 - \bar a z)}{(1 - \bar a z) + \bar a(z - a)} = \frac{z(1 - |a|^2)}{1 - |a|^2} = z.$$

By symmetry ($a \leftrightarrow -a$), $\varphi_a \circ \varphi_{-a} = \mathrm{id}$ too. So φ_a is a bijection of $\mathbb{D}$ with inverse φ_{-a}.

(b) Let f be a conformal bijection of $\mathbb{D}$, and $a = f^{-1}(0)$. Then $g = f \circ \varphi_{-a}$ is a conformal bijection of $\mathbb{D}$ with $g(0) = f(a) = 0$. By Problem 16, $g(z) = e^{i\theta}z$. Hence $f = g \circ \varphi_a = e^{i\theta}\varphi_a$.

23.

f is holomorphic. f is continuous on Ω, as a locally uniform limit of continuous functions. Let D be an open disc with $\overline{D} \subseteq \Omega$ and $T \subseteq D$ a closed triangle. By Goursat, $\int_{\partial T} f_n = 0$. Since ∂T is compact, $f_n \to f$ uniformly on it, so

$$\Big|\int_{\partial T} f\Big| = \Big|\int_{\partial T} (f - f_n)\Big| \le \mathrm{length}(\partial T) \sup_{\partial T} |f - f_n| \to 0.$$

By Morera's theorem, f is holomorphic on D; such discs cover Ω.

Derivatives converge. Let $K \subseteq \Omega$ be compact. Choose $r > 0$ such that $K_r = \{z : \mathrm{dist}(z, K) \le r\} \subseteq \Omega$ (possible because $\mathrm{dist}(K, \mathbb{C} \setminus \Omega) > 0$ when $\Omega \neq \mathbb{C}$; take r less than half of it, or $r = 1$ if

$\Omega = \mathbb{C}$). K_r is closed and bounded, hence compact. For $z \in K$, the circle $|\zeta - z| = r$ lies in K_r, and by the Cauchy integral formula for the derivative,

$$|f_n'(z) - f'(z)| = \left| \frac{1}{2\pi i} \int_{|\zeta - z| = r} \frac{f_n(\zeta) - f(\zeta)}{(\zeta - z)^2} d\zeta \right| \leq \frac{1}{2\pi} \cdot 2\pi r \cdot \frac{\sup_{K_r} |f_n - f|}{r^2} = \frac{\sup_{K_r} |f_n - f|}{r}.$$

The right side is independent of $z \in K$ and tends to 0. So $f_n' \to f'$ uniformly on K.